Luong (Kevin) Tsai

Rowland Heights, CA 91748 | 626-495-7835 | luongqtsai@gmail.com | LinkedIn

Education

California State University, Fullerton **Expected Graduation:** May 2027

B.S. Mechanical Engineering **GPA:** 3.33/4.00

Skills

CAD Software: SOLIDWORKS, AutoCAD, NX 12.0, Fusion 360

Programming Languages: MATLAB, Python, C++

Relevant Coursework: Mechatronics, Fluid Mechanics, Heat Transfer, Design of Thermal and Fluid Systems

Projects

4-DOF Robotic Arm | SOLIDWORKS, ANSYS, Python, DFMA March 2025 - October 2025

- Designed a 4-DOF robotic arm with a gear-driven base, 16cm of reach, and a 45g payload capacity
- Adjusted base hardware to shift the center of mass 20% closer to the axis of rotation, preventing tip-over at max extension
- Implemented inverse kinematics using matrix transformation to control arm movement across all four axes
- Tested maximum reach using incremental weights to find servo stall limits and keep the base from tipping under a heavy load

Autonomous Companion Robot | SOLIDWORKS, Fritzing, Python, Signal Processing, DFMA January 2026 - Present

- Designed a prototype equipped with SG90 servo motors, a touch sensor, and an OLED screen that responds via touch-sensor feedback
- Optimized electrical routing by designing custom mounts for electrical hardware, opening up 40% of inside space
- Eliminated 30% of peak voltage drops to prevent microcontroller brownouts by wiring battery cells to a central power hub
- Developed an asynchronous event loop to synchronize the OLED screen and servo motors, cutting 20% of input latency

Extracurricular Activities

Titan Rover | SOLIDWORKS, FEA, GD&T, CNC Machining, DFMA May 2026 - Present

Robotics subteam

- Designed electronics mounts to free up 15% of inside space, improving wire management and future hardware integration
- Implemented an adjustable roller chain tensioner to reduce rotational backlash in the arm joint by 20%
- Redesigned mounting assembly between the arm and claw to add 2 degrees of freedom to the robotic arm
- Integrated cycloidal gearboxes into the shoulder and elbow joints for a 35% increase in lifting torque under heavy load

3D printed Dental Aligner Night Guards

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